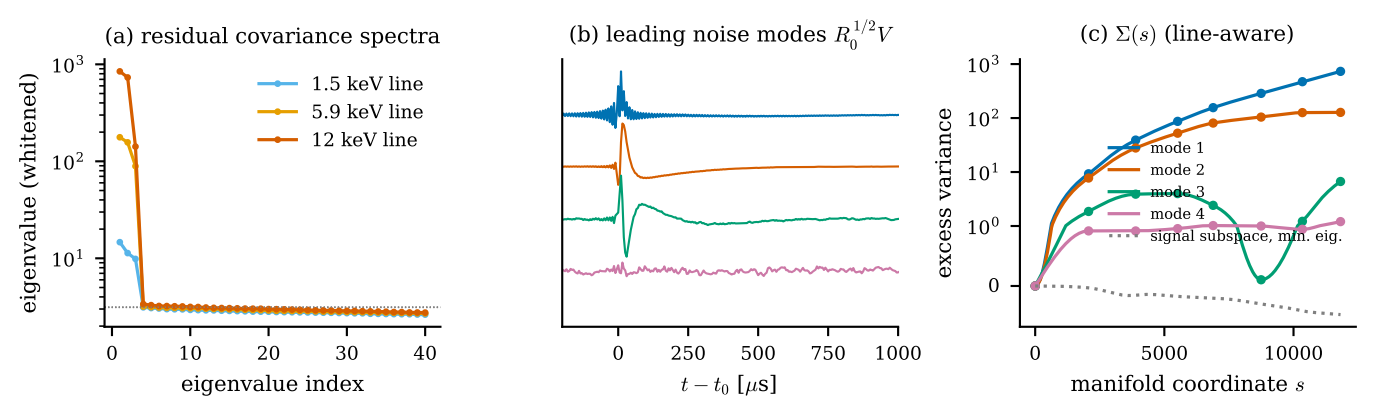
(a) residual covariance spectra
eigenvalue (whitened)
10^3
10^2
10^1
1.5 keV line
5.9 keV line
12 keV line
0
10
20
30
40
eigenvalue index
(b) leading noise modes $R_0^{1/2}V$
0
250
500
750
1000
$t - t_0$ [μs]
(c) $\Sigma(s)$ (line-aware)
excess variance
10^3
10^2
10^1
10^0
0
mode 1
mode 2
mode 3
mode 4
signal subspace, min. eig.
0
5000
10000
manifold coordinate s